generated_report_B: PathMNIST — synthetic images to balance tissue classes

Verdict: Use with restrictions — the synthetic images gave a small, statistically clear gain for a tissue classifier, but they are visibly imperfect, the original imbalance was mild, and the gain cannot be credited to "balancing" specifically.

1. Summary

Question: Can synthetic images top up under-represented tissue types and help a downstream tissue classifier? **Approach:** One class-conditional diffusion model (12 minutes of training), 5,770 synthetic images for the eight smaller classes, compared with plain real data, class re-weighting and random oversampling. **Result:** Macro-F1 on the independent test set was 0.827 with synthetic data vs 0.808 for the best baseline (interval for the difference: +0.015 to +0.023). But a classifier separates synthetic from real 91% of the time, and only 53% of synthetic images were recognised as the intended tissue. **Recommendation:** Use as a supplement only (marked synthetic, real validation/test data), not as a substitute for real images; class weights/oversampling add nothing here.

2. Data at a glance

- 107,180 colon-pathology patches, 28×28 RGB, nine tissue classes; 89,996 for training, 10,004 validation (same source), 7,180 test (different patients and staining).
- The imbalance is mild: the largest training class (adenocarcinoma epithelium, 12,885) is only 1.64× the smallest (normal colon mucosa, 7,886), below the ~3× level where balancing usually matters.
- Neighbouring pixels are strongly correlated (0.90), so image-aware (convolutional) networks are appropriate.

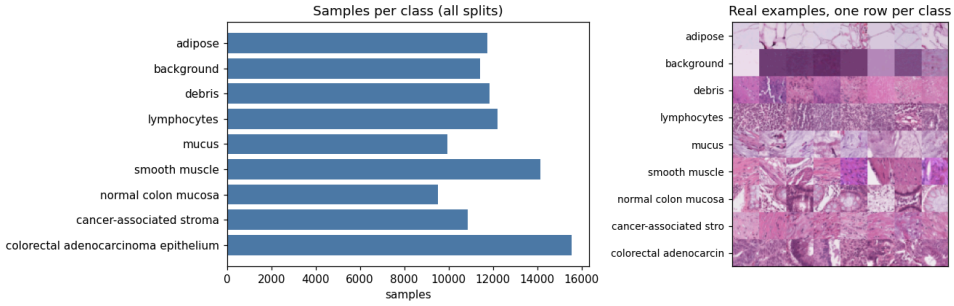


Figure 1: Data profile

3. Decision path

#	Step	Decision	Why
1	Goal	Balancing task; imbalance is mild, likely decline	Largest class is only 1.6 times the smallest; simple fixes probably suffice
2	Data prep	Scale pixels to [−1, 1], keep image shape	Standard input range for image diffusion models; keeps spatial structure
3	Model	Conditional diffusion model with conv network	Plenty of image data and we must choose which tissue to generate
4	Settings	Small diffusion model, 12 minutes training	Probe speed is ~11 steps/s; this fits roughly 8-10k updates in budget
5	Training check	Training looked healthy	Loss fell smoothly and noise prediction improved at every noise level
6	Generate	Top up 8 classes to the largest class	Balancing means raising smaller classes to the size of the biggest
7	Evaluate	Check realism, copying and classifier benefit	Balancing is judged by classifier gain over simple baselines, plus realism
8	Recommend	Use with restrictions; baselines first	Small classifier gain, but images are visibly imperfect and imbalance is mild

The turning point was the first decision: because the imbalance is so mild, the question became whether synthetic data beats simple fixes at all, so the baselines were included in the test from the start.

4. What was generated

5,770 synthetic 28×28 images from a class-conditional diffusion model, only for the eight classes smaller than adenocarcinoma epithelium (between 156 and 1,111 per class, scaled to the 20,000-image real subset used in the classifier test). Real held-out images are shown for comparison. Synthetic images are recognisable but often darker or more saturated than real ones, and the normal-mucosa samples lack the gland structure of real mucosa.

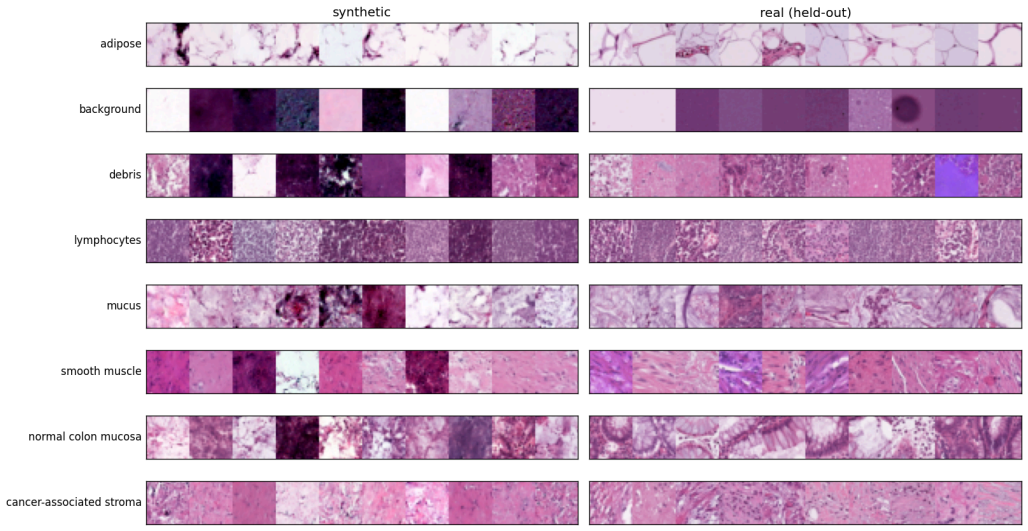


Figure 2: Synthetic (left) vs real held-out (right) images by class

5. Results

Downstream classifier (real test set from another collection; average of 3 runs).

Training data	Accuracy	Macro-F1
Real only	0.859	0.808
Real, class-weighted	0.854	0.799
Real, random oversampling	0.854	0.798
Real + synthetic	0.877	0.827

Gain over the best baseline: +0.019 macro-F1 (interval +0.015 to +0.023), well beyond the run-to-run spread of the baseline (0.003). Eight classes were topped up; average F1 on them rose from 0.793 to 0.816.

Realism and copying (synthetic vs real validation images, same classes).

Measure	Synthetic	Real vs real
Distinguishable from real (classifier accuracy; 0.5 = no)	0.913	0.502
Realism (precision)	0.60	0.96
Coverage (recall)	0.80	0.96
Recognised as intended class	0.53	0.96

Recognition was worst for normal colon mucosa (0.23) and cancer-associated stroma (0.42). There was no sign of copying: the nearest real neighbour of a synthetic image was a training image 50.1% of the time (50% expected), with no exact copies.

6. Recommendation

Use with restrictions. The pre-set criterion (beat the best non-generative baseline with an interval excluding zero) was met, so the synthetic images (`ddpm_conv_ch32/samples_balance.npz`) can supplement classifier training. Restrictions:

- Mark them as synthetic; keep validation and test data real.
- They live in 28×28 pixel space, with images produced at guidance 1.5.
- Label noise: about half of the synthetic images are not recognised as their intended class, so classes such as normal mucosa gain little real information.
- The gain may come from extra data or regularisation rather than balance (see Limitations), and the original imbalance was mild, so class weights or oversampling are not needed.
- No formal privacy guarantee; the copying check is evidence, not proof.

What would change this: a size-matched control (e.g. extra synthetic images of all classes), lower guidance or longer training giving more realistic images, or a more strongly imbalanced real dataset.

7. Limitations

- No control isolates balancing from simply adding more (synthetic) data; the real training subset used was 20,000 images.
- The classifier is small and trained for 6 epochs on 3 seeds; the interval covers test-set noise only.
- The realism judge is a small CNN trained on the same data, not a standard pretrained network.
- Training was limited to 12 minutes (about 15.6k steps, 40,000-image subsample); no tuning of guidance or sampling was done, and no second model family was compared.
- The test set comes from a different source, so absolute scores also reflect a source shift.

Appendix A. Full decision log

1. **Goal** *(objective skill)* — Task = class balancing. Success: recommend synthetic only if real+synthetic macro-F1 beats best of real_only/class-weighted/oversample with paired bootstrap CI excluding 0 and diff > 2 seed SDs; else recommend baseline. Real reference: val for fidelity, test for classifier. Conditioning on y needed. Run one small conditional DDPM to test this empirically since the augment evaluation supplies the baselines.
   - *Evidence:* profile.json: imbalance_ratio_max_over_min=1.64 (<3), smallest class normal colon mucosa n_train=7886, normalized_entropy=0.9949
   - *Alternatives rejected:* Decline outright without evidence: rejected, baselines come with augment evaluation at modest cost
2. **Data prep** *(preprocessing skill)* — –prep scale_pm1, conv backbone, no flattening
   - *Evidence:* profile.json: value_range 0-255, neighbour_pixel_correlation=0.904, no missing values
   - *Alternatives rejected:* mlp/flatten: discards spatial structure
3. **Model** *(model-selection skill)* — conditional DDPM, conv UNet, ch=32; no second family
   - *Evidence:* n_train=89996 >> 5k; image data; need class control (model-selection rule 3)
   - *Alternatives rejected:* cVAE: blurry samples; WGAN-GP: fragile, no specific reason
4. **Settings** *(diffusion skill)* — ch=32, T=1000 cosine, p_uncond=0.1, ema 0.999, lr 2e-4, batch 64, max-train 40000 (stratified), max-minutes 12 (probe used 1; total 13 of 15)
   - *Evidence:* probe train_log: 156 steps in 14.3s on mps; smoke test passed
   - *Alternatives rejected:* ch=64: ~4x slower, too few steps; full 90k train: no benefit within the step budget

5. **Training check** *(diffusion skill)* — Health check passed; stopped by 12-min limit (25 epochs, ~15.6k steps at batch 64)
   - *Evidence:* train_log.json: eps_mse t5=0.229, t25=0.062, t50=0.020, t75=0.0056, t95=0.0007; loss.png plateau ~0.056 without divergence
   - *Alternatives rejected:* none needed
6. **Generate** *(diffusion skill)* — class-counts (20k-subset scale) 782/750/561/552/1084/156/1111/774 for classes 0-7, none for class 8; DDIM 50 steps, guidance 1.5, tag balance; 5770 samples
   - *Evidence:* sample_config_balance.json: 164.2s, non-finite 0; targets scaled from profile.json class_counts_train to the 20,000-image real subset used by the augment test
   - *Alternatives rejected:* guidance 1.0: less class-typical; more samples: no budget
7. **Evaluate** *(evaluation skill)* — Ran fidelity, memorization and augment utility on the balance samples. Criterion met nominally: real+synthetic macro-F1 0.8271 vs best baseline (real_only) 0.8080; diff CI [0.0145,0.0232]; seed SD 0.0007-0.018. But fidelity weak: C2ST 0.913 (ref 0.5015), precision 0.603 / recall 0.802 (ref 0.964/0.962), label consistency 0.5265 (ref 0.9636), mucosa 0.2313. No copying: nearest-train fraction 0.501 (expected 0.5), 0 exact copies. Caveat: no size-matched control, so the gain cannot be attributed to balancing; the real subset was already near balanced (1.64x).
   - *Evidence:* ddpm_conv_ch32/metrics_balance.json
   - *Alternatives rejected:* No further iteration: budget nearly spent (13/15 min); lower guidance resample would not remove the missing size-matched control
8. **Recommend** *(recommendation skill)* — Use with restrictions: samples_balance.npz may supplement training of the tissue classifier (flag as synthetic; keep val/test real); validate on real held-out data from a different source; do not use as a stand-in for real images. Class weights/oversampling are not needed for this mild imbalance and were not better than plain real data.
   - *Evidence:* metrics_balance.json: macro-F1 0.8271 vs 0.8080/0.7988/0.7983 (real_only/class-weighted/oversample); profile.json imbalance 1.64
   - *Alternatives rejected:* Do not use: gain excludes 0 and no copying; Use: fidelity too weak

Appendix B. Training details

- Conditional diffusion model (small convolutional UNet, 448k parameters in the probe configuration, width 32), 1,000 noise steps with cosine schedule, label dropout 0.1 for guidance, EMA 0.999, learning rate 2e-4, batch 64, 40,000-image stratified training subset.
- Stopped by the 12-minute limit after 25 epochs (721 s); a 1-minute probe used 1 further minute (13 of 15 minutes in total).
- Health check: loss plateaued near 0.056 without divergence; noise-prediction error fell from 0.229 (low noise) to 0.0007 (high noise).
- Sampling: 50 DDIM steps, guidance 1.5, 164 s for 5,770 images.

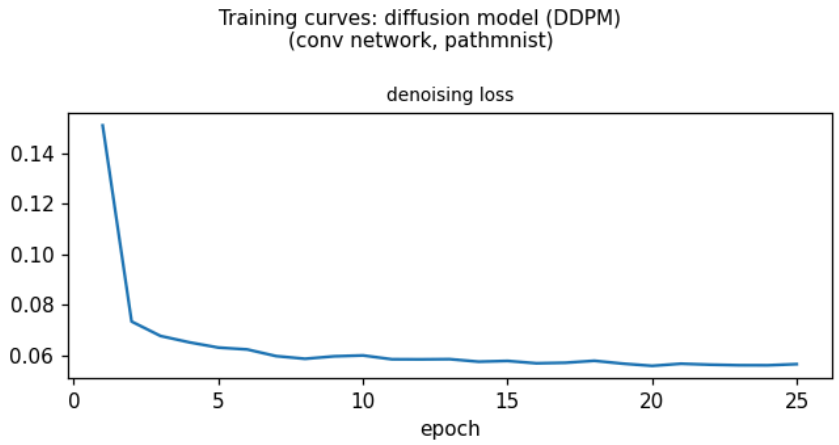


Figure 3: Training loss

Appendix C. Glossary

- **Macro-F1:** F1 score averaged equally over classes; sensitive to weak classes.
- **Class weighting / oversampling:** non-generative ways of giving small classes more influence in training.
- **Diffusion model (DDPM):** generates images by gradually removing noise; here conditioned on tissue class.
- **Guidance:** a setting that makes generated images more typical of the requested class, at the cost of variety.
- **Precision / recall (of synthetic data):** share of synthetic images that look real / share of real variety covered.
- **Classifier two-sample test:** how well a classifier tells synthetic from real; 0.5 means it cannot.
- **Label consistency:** share of synthetic images a real-trained classifier assigns to the intended class.
- **Paired bootstrap interval:** uncertainty range for a difference in scores, from resampling the test set.

Appendix D. Reproduce

Tool calls in the order the agent ran them (smoke tests and –help omitted):

```
python3 tools/inspect_data.py --data data/pathmnist --out outputs/
generated_report_B_pathmnist
python3 tools/train.py --data data/pathmnist --out outputs/
generated_report_B_pathmnist/ddpm_probe --model ddpm --backbone conv --prep scale_pm1
--conditional --max-train 20000 --epochs 1 --batch-size 128 --lr 2e-4 --
params '{"ch":32,"T":1000,"schedule":"cosine","p_uncond":0.1,"ema_decay":0.999}' --
max-minutes 1
python3 tools/train.py --data data/pathmnist --out outputs/
generated_report_B_pathmnist/ddpm_conv_ch32 --model ddpm --backbone conv --prep
scale_pm1 --conditional --max-train 40000 --epochs 200 --batch-size 64 --lr 2e-4
--params '{"ch":32,"T":1000,"schedule":"cosine","p_uncond":0.1,"ema_decay":0.999}' --
max-minutes 12
python3 tools/sample.py --run outputs/
generated_report_B_pathmnist/ddpm_conv_ch32 --class-counts
'{"0":782,"1":750,"2":561,"3":552,"4":1084,"5":156,"6":1111,"7":774,"8":0}' --steps 50
```

```
--guidance 1.5 --tag balance
python3 tools/evaluate.py --run outputs/generated_report_B_pathmnist/ddpm_conv_ch32 --
tasks fidelity,memorization,utility --utility augment --samples-tag balance
```